

Project Plan

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1 Research background

Gravitational waves (GW) produced by merging binary black holes (BBH) can be lensed by massive objects such as galaxies along their path, potentially leading to observation of multiple images with similar waveform as the GW take different paths to the detectors. Lensed signals are not exactly identical as each image passes through a different part of the lens and hence experiences different magnification. Different arrival times of the images and therefore different detector noises also contribute to the difference in waveform. On the other hand, GW signals originating from two individual merger events of two pairs of BBH with approximately the same parameters may look like copies of one another. Thus, we have to find a way to determine whether similar signals are gravitationally lensed or unlensed.

There is no strong evidence demonstrating gravitational lensing yet. As light is lensed by massive objects in a similar way as GW, we may deduce the BBH source population and lens population from the paths of light, which affect the probability of lensing. Consequently, addition of electromagnetic (EM) information may increase our confidence in determining whether two similar signals are from two pairs of very similar BBH or from one pair of merging BBH lensed.

Mathematically, according to Bayes' theorem, combining prior knowledge (GW signals) with new evidence (EM information) gives rise to updated, revised belief which is more accurate.

2 Research challenge

The main goal is to confirm whether the addition of EM information to GW signals would provide stronger evidence of strong gravitational lensing. The overall project is divided into two branches: the GW branch, which aims to identify and quantify evidence for strong lensing using GW observations alone, and an EM branch, which reconstructs the lensing galaxies from EM data to provide independent constraints that can strengthen the GW-based lensing claims.

In the GW branch, GW signals are first simulated and analysed through parameter estimation, giving posterior samples. With the use of FIGARO, these are then converted to BBH population density, which becomes the updated prior ($P(\theta)$) in the

Bayes factor. Posterior densities, which are the probability densities of parameters conditioned on certain observed data ($P(\theta|d)$), can also be computed from the posterior samples. The Bayes factors comparing the lensed and unlensed hypotheses (B_U^L) of lensing candidates are then computed.

In parallel, the EM branch begins with the construction of an electromagnetic lens population, which is prepared in advance. The lensing population is then reconstructed using gwemfish.

Finally, the results from the GW and EM branches are combined to determine how much multimessenger (MM) information strengthens the evidence for strong lensing, specifically by assessing whether a candidate with moderate significance (e.g. around 2σ) based on GW data alone can be promoted to a highly significant detection (e.g. 5σ) when EM information is included.

3 Steps to solve research challenge

Figure 1 shows the workflow of the GW branch.

The Bayes factor comparing the lensed and unlensed hypotheses can be found with

$$B_U^L = \frac{P(d_1, d_2 | \text{lensed})}{P(d_1, d_2 | \text{unlensed})} = \int \frac{P(\theta | d_1) P(\theta | d_2)}{P(\theta)} d\theta,$$

where θ is the set of BBH parameters $m_1, m_2, \dots$ and d_1 and d_2 are the data associated with events 1 and 2. The 15-dimensional integral can be simplified using importance sampling to save computational cost:

$$B_U^L \approx \frac{1}{N} \sum_{i=1}^N \frac{P(\theta_i | d_2)}{P(\theta_i)}, \quad \theta_i \sim P(\theta | d_1),$$

where N is the number of posterior samples and θ_i is the i -th sample drawn from $P(\theta | d_1)$.

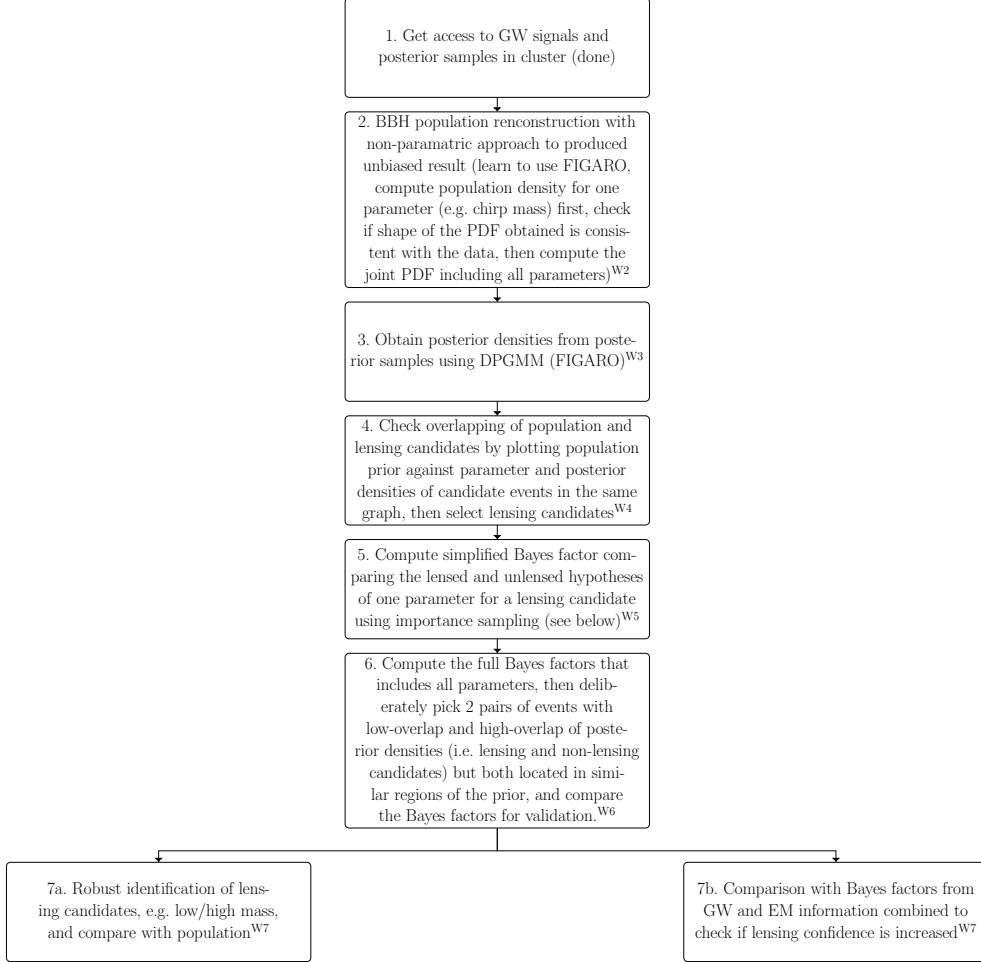


Fig. 1 workflow of solving research challenge (GW branch).